

CYBER -ATTACK DETECTION LAYER

Introduction:

The proposed framework employs a multi-layer detection and mitigation strategy to identify and respond to the cyber-attacks modelled in chapter 3. Each layer targets a specific class of attack and compensates for the limitations of other methods. The economic operation of the proposed Cyber-Physical Power System (CPPS) is governed by a decentralized application (DApp) built on the Ethereum blockchain. This layer facilitates peer-to-peer (P2P) energy trading among prosumers while ensuring transparency, security, and tamper-proof execution of transactions. By tightly integrating with the detection and mitigation framework, the market layer acts as both a financial settlement engine and a security enforcement mechanism.

Weighted Least Squares (WLS) State Estimation:

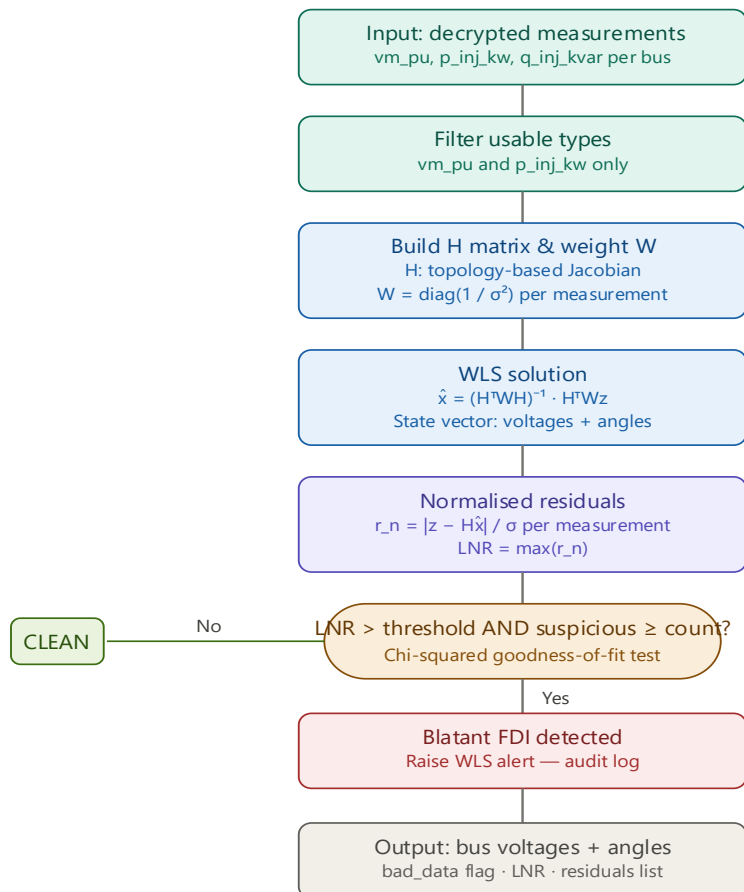


Figure: WLS flowchart

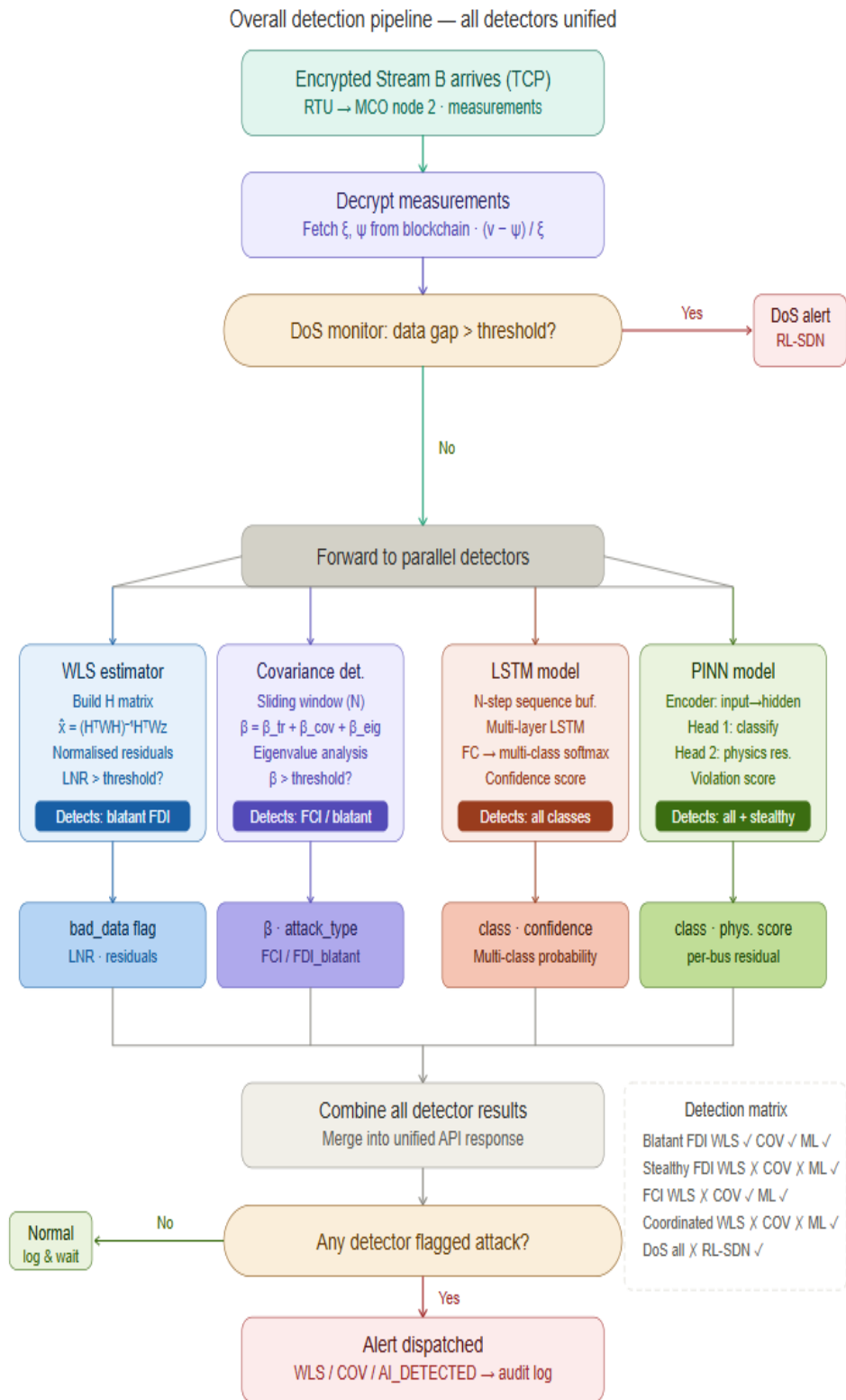


Figure: detection pipeline for the cpps model

The detection layer forms the intelligent core of the proposed CPPS framework. It receives the decrypted measurement vector from the communication layer and passes it through a four-stage pipeline in which each stage targets attack classes that prior stages cannot reliably identify. The first stage applies Weighted Least Squares (WLS) state estimation to detect blatant FDI attacks through normalised residual testing. The second stage employs the Chakrabarty-Sikdar covariance anomaly detector to identify FCI attacks and coordinated multi-bus FDI through off-diagonal correlation structure in the measurement covariance matrix.

The third stage uses a two-layer LSTM network that processes temporal sequences of measurement vectors to detect stealthy FDI attacks that lie in the null space of the Jacobian and are therefore invisible to both WLS and the covariance detector. The fourth stage employs a Physics-Informed Neural Network with a dual-head architecture that simultaneously classifies the attack type and computes per-bus physics residuals, providing spatial localisation of the affected buses. For Denial-of-Service attacks on the communication channel, a Q-learning Reinforcement Learning agent coupled with an SDN controller monitors data-arrival continuity and autonomously reroutes measurement traffic across four redundant paths upon detecting a sustained data gap